

A BAYESIAN ANALYSIS OF THE MULTIVARIATE CONTROLLED CALIBRATION PROBLEM

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curves; posterior distribution.

Summary: In this paper the Bayesian approach of Hunter and
Lamboy (1981) is extended from univariate to multi-
variate calibration. Firstly, the known covariance
case will be considered. Although it is difficult to
derive the exact posterior distribution of the unknown
controlled variable ξ , the exact moments can be ob-
tained. By calculating the first four moments for the
paint data given in Brown (1982) and by using Pear-
son curves or Cornish-Fisher expansions, it will be
shown that good approximations of the true posterior
distribution can be obtained.

For the unknown variance case a simulation study
was conducted. From this it became apparent that
Hunter and Lamboy's approach, extended to multi-
variate calibration, does not differ much from the
multivariate extension of the classical univariate
estimator.

1. INTRODUCTION

Since Krutchkoff's paper appeared in *Technometrics* (1967), a great deal has
been written on the problem of calibrating measurement instruments. The
mean-squared errors of the classical and the inverse estimators have been
studied by Krutchkoff (1967), Lwin and Maritz (1982, Randall (1985) and
Oman (1985). Conditions under which one estimate is superior to another

and vice versa are also obtained from these papers. Hunter and Lamboy (1981) presented another approach to this problem. This Bayesian solution in some sense justified the "classical" estimator in much the same way as Hoadley's (1970) and Brown's (1982) methods justified the "inverse" estimator.

In this paper the Bayesian approach of Hunter and Lamboy will be extended from univariate to multivariate calibration. The motivation of this research originated from the question raised by Prof. Dunsmore (in discussion of Brown's (1982) paper), namely whether it had been investigated if the classical procedure used by Brown (1982) will differ much (from a practical point of view) from Hunter and Lamboy's approach in the multivariate case. For further details about the classical procedure in the multivariate case see Lieftinck-Koeijers (1988), Nishii and Krishnarajah (1988), and Clarke (1992) for calibration in multivariate non-linear regression.

Mathematically the multivariate calibration procedure can be formulated as follows. Let the true values associated with a set of p responses and a set of q controlled variables be designated by $\eta = (\eta_1, \dots, \eta_p)'$ and $\xi = (\xi_1, \dots, \xi_q)'$ respectively. Since it was suggested by Brown (1982) that one should predict ξ one variable or factor at a time, only the special case $q = 1$ will be considered in this study. Assume further that the elements of the response vector η are linearly related to ξ , i.e.

$$(1) \quad \eta = \beta_0 + \beta_1 \xi$$

where $\beta_0 = (\beta_{01}, \beta_{02}, \dots, \beta_{0p})'$ and $\beta_1 = (\beta_{11}, \beta_{12}, \dots, \beta_{1p})'$.

Consider an initial sample (x_i, y_i) , $i = 1, \dots, n$ which will be called the calibration sample. The x_i and y_i ($p \times 1$) are the observed values of ξ_i and η_i respectively,

$$(1.2) \quad y_i = \eta_i + \epsilon_i \quad i = 1, \dots, n$$

$$x_i = \xi_i + \delta_i \quad i = 1, \dots, n$$

where ϵ_i ($p \times 1$) and δ_i are experimental errors. It is a useful approximation in many situations to assume $\delta_i = 0$ for all i , for example when the variance of δ is much smaller than that of ϵ as is usually the case in practice. Making this assumption and substituting (1.2) in (1.1), it follows that

$$(1.3) \quad y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

The ϵ_i ($i=1, \dots, n$) are assumed to be independently normally distributed with mean 0 and common covariance matrix Σ .

The calibration equation (1.3) is determined by the unknown parameters β_0 and β_1 . As in Hunter and Lamboy (1981) a Bayesian procedure will be used to make inferences about these parameters.

Having established the calibration equation, it is now possible to proceed to the second stage. A new observation is presented with a specific unknown value η , and a measurement

$$(1.4) \quad y_0 = \eta + \epsilon_0$$

is obtained where $y_0 = (y_{01}, y_{02}, \dots, y_{0p})'$. It is also assumed that ϵ_0 has the same distribution as the errors in the first stage. In the univariate case ($p=1$) Hunter and Lamboy (1981) derived the posterior distribution of $\xi = (\eta - \beta_0)/\beta_1$ from which inferences about the unknown ξ were made.

In the next section the work of Hunter and Lamboy (1981) will be extended to multivariate calibration and a Bayes solution will be presented for the calibration problem on the assumption that Σ is known. Although, it is difficult to derive the exact posterior distribution of ξ , the exact moments can be obtained. By calculating the first four moments for the paint data given in Brown (1982, p. 301) and by using Pearson curves or Cornish-Fisher expansions it is shown in section four that good approximations of the true posterior distribution can be obtained.

In the last section the unknown-variance case is considered. By conducting a simulation study it appears that the results of our procedure do not differ much (from a practical point of view) from the classical approach.

2. A BAYESIAN SOLUTION - Σ KNOWN

Given the data from both the first and the second stages and by assuming locally uniform priors on β_0 , β_1 and η it can easily be shown (see for example Hunter and Lamboy's equation (2.1) for the univariate case) that the joint posterior distribution $p(\beta_0, \beta_1, \eta | \text{data})$ is normal with mean $(\hat{\beta}_0', \hat{\beta}_1', \hat{\eta}')'$ and covariance matrix

$$\hat{S} = \begin{bmatrix} (X'X)^{-1} \otimes \Sigma & 0 \\ 0 & \Sigma \end{bmatrix} = \begin{bmatrix} \bar{A} \Sigma & \bar{B} \Sigma & 0 \\ \bar{B} \Sigma & \bar{C} \Sigma & 0 \\ 0 & 0 & \Sigma \end{bmatrix}$$

where $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$ are the usual least squares

estimates of β_0 and β_1 . Furthermore $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and X is the $n \times 2$ matrix, the i -th row of which is $(1, x_i)$ which means that

$$(X'X)^{-1} = \begin{bmatrix} \bar{A} & \bar{B} \\ \bar{B} & \bar{C} \end{bmatrix} \quad \text{where} \quad \bar{A} = \frac{\sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2}, \quad \bar{B} = \frac{-\bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{and}$$

$$\bar{C} = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}. \quad \text{Also } \otimes \text{ is the usual Kronecker product sign. Note that } \eta$$

is independently distributed of β_0 and β_1 since the observation taken at the second stage is independent of those taken at the first stage.

From the joint posterior distribution $p(\beta_0, \beta_1, \eta | \text{data})$ it follows that the posterior distribution $((\eta - \beta_0)', \beta_1')'$ is normal with mean $((\bar{y}_0 - \hat{\beta}_0)', \hat{\beta}_1')'$

and covariance matrix $\begin{bmatrix} (A+1)\Sigma & -B\Sigma \\ -B\Sigma & C\Sigma \end{bmatrix}$. Further the conditional distribution of $\eta - \beta_0$ given β_1 is also normal with mean $(\bar{y}_0 - \bar{y}) + \bar{x}\beta_1$ and

not to be confused with the usual covariance matrix $(1 + \frac{1}{n})\Sigma$.

In the univariate case ($p = 1$), Hunter and Lamboy (1981) derived the posterior distribution of ξ .

$$(2.1) \quad \xi = (\eta - \beta_0) / \beta_1$$

from which inferences about the unknown ξ were made.

A natural analog to (2.1) in the case of multivariate calibration is

$$(2.2) \quad \xi = \frac{(\eta - \beta_0)' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1}$$

which is obtained by minimizing

$$(2.3) \quad (\eta - \beta_0 - \beta_1 \xi)' \Sigma^{-1} (\eta - \beta_0 - \beta_1 \xi)$$

with respect to ξ .

Equation (2.2) resembles the multivariate extension of the classical univariate estimator

$$\xi = \frac{(\eta - \beta_0)' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1}$$

considered by Liefstinck-Koeijers (1988) and Brown (1982).

Since equation (2.1) is the ratio of two normal variates it does not possess finite moments and has infinite risk relative to quadratic and other loss functions. This disturbing feature does not occur in (2.2). In the next section it will be shown that the mean, variance and third moment of ξ exist for $p > 2$ and the fourth moment for $p > 4$.

3. EXACT MOMENTS OF $\xi = \frac{(\eta - \beta_0)' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1}$

The exact distribution of ξ is not known but by using the first four moments of ξ and Pearson curves or Cornish-Fisher expansions, good approximations of the exact distribution of ξ can be obtained. Therefore it

is the purpose of this section to derive the first four moments of ξ for Σ known.

In order to obtain the moments of ξ , results in Ullah (1974) and Egerton and Laycock (1982) will be used. The following theorem can now be proved.

THEOREM 3.1

The first four moments of $\xi = \frac{(\eta - \beta_0)' \Sigma^{-1} \beta_1}{(\beta_1' \Sigma^{-1} \beta_1)}$ are given by

$$(3.1) \quad E(\xi) = C^{\frac{1}{2}} \mathbf{h}' \tau E\left(\frac{1}{2K+p}\right) + \bar{x}, \quad (3.1)$$

$$(3.2) \quad E(\xi^2) = C\left(1 + \frac{1}{n}\right) E\left(\frac{1}{2K+p-2}\right) + C\left\{\frac{1}{2}(\mathbf{h}' \mathbf{h}) E\left(\frac{1}{2K+p-2}\right) + \left(\frac{1}{2}(\mathbf{h}' \tau)^2 - \frac{1}{2}(\mathbf{h}' \mathbf{h})\right) E\left(\frac{1}{2K+p}\right) - \frac{1}{2}(\mathbf{h}' \tau)^2 E\left(\frac{1}{2K+p+2}\right)\right\} + 2\bar{x}C^{\frac{1}{2}} \mathbf{h}' \tau E\left(\frac{1}{2K+p}\right) + \bar{x}^2$$

$$(3.3) \quad E(\xi^3) = \left\{\frac{1}{8}C^{\frac{3}{2}}(\mathbf{h}' \tau)^3 E\left(\frac{1}{2K+p+4}\right) + C^{\frac{3}{2}}\left[\frac{2}{8}(\mathbf{h}' \tau)^3 - \frac{3}{8}(\mathbf{h}' \mathbf{h})(\mathbf{h}' \tau)\right] - \frac{3}{2}\bar{x}C(\mathbf{h}' \tau)^2\right\} E\left(\frac{1}{2K+p+2}\right) + \left\{-\frac{3}{2}C^{\frac{3}{2}}\left(1 + \frac{1}{n}\right)(\mathbf{h}' \tau) + \frac{1}{8}C^{\frac{3}{2}}(\mathbf{h}' \tau)^3 - \frac{6}{8}C^{\frac{3}{2}}(\mathbf{h}' \mathbf{h})(\mathbf{h}' \tau) + 3\bar{x}\left[\frac{1}{2}(\mathbf{h}' \tau)^2 - \frac{1}{2}(\mathbf{h}' \mathbf{h})\right] + 3\bar{x}C(\mathbf{h}' \tau)\right\} E\left(\frac{1}{2K+p}\right) + \left\{\frac{3}{2}\left(1 + \frac{1}{n}\right)C^{\frac{3}{2}}(\mathbf{h}' \tau) + 3\left(1 + \frac{1}{n}\right)\bar{x}C + \frac{3}{8}C^{\frac{3}{2}}(\mathbf{h}' \mathbf{h})(\mathbf{h}' \tau) + \frac{3}{2}\bar{x}C(\mathbf{h}' \mathbf{h})\right\} E\left(\frac{1}{2K+p-2}\right) + \bar{x}^3$$

$$(3.4) \quad E(\xi^4) = \left\{\frac{3}{2}C\left(1 + \frac{1}{n}\right)^2 + \frac{6}{8}C\left(1 + \frac{1}{n}\right)(\mathbf{h}' \mathbf{h}) + \frac{3}{48}C(\mathbf{h}' \mathbf{h})^2\right\} E\left(\frac{1}{2K+p-4}\right) + \left\{-\frac{3}{2}C\left(1 + \frac{1}{n}\right)^2 + \frac{6}{8}C\left(1 + \frac{1}{n}\right)[(\mathbf{h}' \tau)^2 - 2(\mathbf{h}' \mathbf{h})] + 6\bar{x}^2C\left(1 + \frac{1}{n}\right) + \frac{1}{48}C[6(\mathbf{h}' \mathbf{h})(\mathbf{h}' \tau)^2 - 9(\mathbf{h}' \mathbf{h})^2]\right\} + \frac{12}{8}\bar{x}C^{\frac{3}{2}}(\mathbf{h}' \mathbf{h})(\mathbf{h}' \tau) + 3\bar{x}^2C(\mathbf{h}' \mathbf{h})\right\} E\left(\frac{1}{2K+p-2}\right) + \frac{6}{8}C^2\left(1 + \frac{1}{n}\right)[(\mathbf{h}' \mathbf{h}) - 2(\mathbf{h}' \tau)^2] - 12\bar{x}C^{\frac{3}{2}}\left(1 + \frac{1}{n}\right)(\mathbf{h}' \tau)$$

$$\begin{aligned}
 & + \left\{ \frac{1}{48} C^2 [(\mathbf{h}' \boldsymbol{\tau})^4 - 18(\mathbf{h}' \mathbf{h})(\mathbf{h}' \boldsymbol{\tau})^2 + 9(\mathbf{h}' \mathbf{h})^2] + 4\bar{x} C^{\frac{3}{2}} \left[\frac{1}{8} (\mathbf{h}' \boldsymbol{\tau})^2 \right. \right. \\
 & - \frac{6}{8} (\mathbf{h}' \mathbf{h})(\mathbf{h}' \boldsymbol{\tau}) + 6\bar{x} C \left[\frac{1}{2} (\mathbf{h}' \boldsymbol{\tau})^2 - \frac{1}{2} (\mathbf{h}' \mathbf{h}) \right] + 4\bar{x}^3 C^{\frac{1}{2}} (\mathbf{h}' \boldsymbol{\tau}) \left. \right\} E\left(\frac{1}{2K+p}\right) \\
 & + \left\{ \frac{6}{8} C^2 \left(1 + \frac{1}{n}\right) (\mathbf{h}' \boldsymbol{\tau})^2 + \bar{x} C^{\frac{3}{2}} \left(1 + \frac{1}{n}\right) (\mathbf{h}' \boldsymbol{\tau}) - \frac{1}{48} C^2 [3(\mathbf{h}' \boldsymbol{\tau})^4 \right. \\
 & - 18(\mathbf{h}' \mathbf{h})(\mathbf{h}' \boldsymbol{\tau})^2 + 3(\mathbf{h}' \mathbf{h})^2] - 4\bar{x} C^{\frac{3}{2}} \left[\frac{2}{8} (\mathbf{h}' \boldsymbol{\tau})^3 - \frac{3}{8} (\mathbf{h}' \mathbf{h})(\mathbf{h}' \boldsymbol{\tau}) \right] \\
 & - 3\bar{x}^2 C (\mathbf{h}' \boldsymbol{\tau})^2 \left. \right\} E\left(\frac{1}{2K+p+2}\right) \\
 & + \left\{ \frac{1}{48} C^2 [3(\mathbf{h}' \boldsymbol{\tau})^4 - 6(\mathbf{h}' \mathbf{h})(\mathbf{h}' \boldsymbol{\tau})^2] + \frac{1}{2} \bar{x} C^{\frac{3}{2}} (\mathbf{h}' \boldsymbol{\tau})^3 \right\} E\left(\frac{1}{2K+p+4}\right) \\
 & - \frac{1}{48} C^2 (\mathbf{h}' \boldsymbol{\tau})^4 E\left(\frac{1}{2K+p+6}\right) + \bar{x}^4
 \end{aligned}$$

where K is a Poisson random variable with parameter $\tilde{\lambda} = \frac{1}{2} \lambda = \frac{1}{2} \boldsymbol{\tau}' \boldsymbol{\tau}$ with $\boldsymbol{\tau} = C^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \hat{\beta}_1$, $\mathbf{h}' = (\mathbf{y}_0 - \bar{\mathbf{y}})' \Sigma^{-\frac{1}{2}}$ and $C = \bar{C}^{-1} = \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^2$.

PROOF:

The proof for $E(\xi^2)$ is given in the Appendix. The proofs for the other moments follow in the same way after some lengthy algebra. Also the lemmas used in the proof of theorem 3.1 are given in the appendix.

As mentioned Lieftinck-Koeijers (1988) and Brown (1982) considered the estimator

$$(3.5) \quad \hat{\xi} = \frac{(\mathbf{y}_0 - \hat{\beta}_0)' \Sigma^{-1} \hat{\beta}_1}{\hat{\beta}_1' \Sigma^{-1} \hat{\beta}_1}$$

of ξ . The former derived expressions for the mean and variance of $\hat{\xi}$. The expression for the mean is given by

$$(3.6) \quad E(\hat{\xi}) = C(\xi - \bar{x}) \beta_1' \Sigma^{-1} \beta_1 E\left(\frac{1}{p+2K}\right) + \bar{x}$$

where $K \sim \text{Poisson}(C\beta_1' \Sigma^{-1} \beta_1)$. The expression for $E(\xi)$ (equation (3.1)) is similar to that of $E(\hat{\xi})$ and can be obtained from the latter by substituting

$(y_0 - \bar{y})$ for $(x_0 - \bar{x})\beta_1$ and β_1 for β_1 . These similarities in the expected values seem to be an indication that the classical estimator will not differ much from Hunter and Lamboy's "extended" procedure. The derivations given by Lieftinck-Koeijers follow the lines of the proof of lemma 11.3 as carried out by Arnold (1981) but are very difficult to extend especially for obtaining higher moments. We therefore consider our methods more transparent and easier to apply especially for the calculation of higher moments of ξ .

4. APPROXIMATIONS OF THE POSTERIOR DISTRIBUTION OF ξ - Σ KNOWN

As mentioned in the introductory paragraph good approximations of the exact posterior distribution of ξ can be obtained by calculating the first four moments and using Pearson curves or Cornish-Fisher expansions.

For details on how to determine the parameters of a Pearson curve, given the values of its moments, see for example Elderton (1953) or Elderton and Johnson (1969). The Cornish-Fisher approximation is described in Cornish and Fisher (1937) and Fisher and Cornish (1960). For further details see Du Plessis and Van der Merwe (1991).

EXAMPLE 4.1

To apply these approximations, the paint data of the paper of Brown (1982, p. 301) was reused. The same subset of 27 observations was used for calibration. In contrast to Brown's example (1982), where the six responses were reduced to two, all six responses will be used to predict a single factor, namely viscosity. A linear model was fitted to these data and the least squares estimates are given in Du Plessis and Van der Merwe (1991).

As mentioned in section 1, the known Σ case will be considered first. For comparison purposes the sample covariance matrix will be used as Σ .

Prediction of ξ is based on the six responses

$$y_0 = [1,850 \quad 31,50 \quad 68,50 \quad 37,17 \quad 91,50 \quad 22,50]'$$

where y_0 is observation 30 of the original 36 observations and corresponds to a true ξ -value of 1. Also $\bar{y} = [1,75 \quad 29,86 \quad 69,93 \quad 37,93 \quad 94,48 \quad 20,33]'$ and $\bar{x} = 1$.

By using equations (3.1) – (3.4) the first four moments of ξ as well as the moments about the mean and the coefficients of skewness and kurtosis, $\tilde{\beta}_1$ and $\tilde{\beta}_2$ have been calculated. From these statistics it was clear that a Type IV Pearson curve is appropriate for this example.

To assess the usefulness of the Pearson curve and Cornish-Fisher approximations, a simulation study of 10 000 observations has been conducted, representing the "exact" distribution of ξ . This was done in the following way:

- (i) Draw β_1 from a $N(\hat{\beta}_1, \hat{\Sigma})$.
- (ii) Given β_1 draw $(\eta - \beta_0)$ from a $N\{(y_0 - \bar{y}) + x\beta_1; (1 + \frac{1}{n})\Sigma\}$,
- (iii) Calculate $\xi = \frac{(\eta - \beta_0)' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1}$.

Repetition of steps (i) – (iii) yields a fair representation of the posterior distribution of ξ .

The histogram of 10 000 simulated values of ξ as well as the Pearson curve and the Cornish-Fisher approximations are shown in figure 4.1.

From figure 4.1 it is clear that the Pearson and Cornish-Fisher densities fit the "exact" distribution quite well – in fact there is not much to choose between the accuracy of the two approximations.

The 90%, 95% and 99% Pearson curve and Cornish-Fisher approximation credibility intervals as well as the "exact" credibility intervals (obtained from the simulation study) are given in the table below.

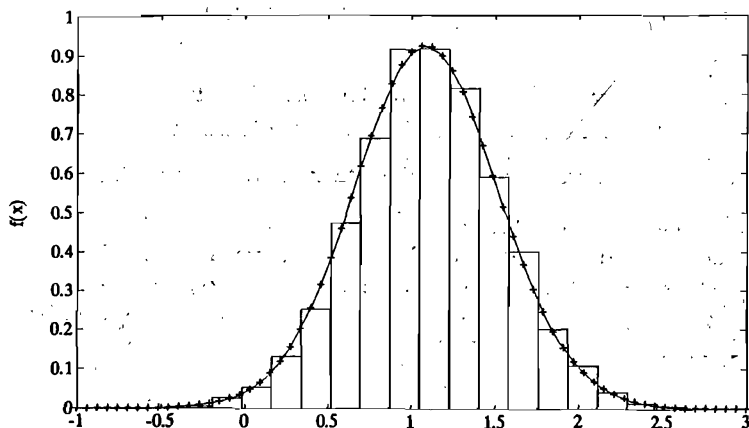
Posterior distribution of $\xi - \Sigma$ known

Figure 4.1

Histogram of 10 000 simulated values of ξ
 — Pearson Curve Approximation
 ***** Cornish-Fisher Approximation

Table 4.2 Credibility intervals

	90%	95%	99%
Pearson	0,3719-1,8110	0,2317-1,9562	-0,0481-2,4898
Cornish-Fisher	0,3719-1,8113	0,2317-1,9568	-0,0490-2,2507
"Exact"	0,3843-1,8130	0,2258-1,9574	-0,0736-2,2501

It was mentioned by one of the referees that $p(\xi)$ looks remarkably like a normal curve. The normal 90% credibility interval: 0,3668-1,8084 is fairly close to the others. The reason for this is that asymptotically most posterior densities tend to be normal.

For further details on this example see Du Plessis and Van der Merwe (1991).

5. THE UNKNOWN VARIANCE CASE

We assume that little is known, a priori, about the parameters, the $p(p+1)/2$ distinct elements of Σ . As our diffuse prior probability density function, we assume that the elements of $B = (\beta'_0, \beta'_1)'$ and those of Σ are independently distributed: that is

$$(5.1) \quad p(B, \Sigma) = p(B)p(\Sigma).$$

Using the invariance theory due to Jeffreys (1961), we take as mentioned in section 2

$$(5.2) \quad p(B) = \text{constant}$$

and

$$(5.3) \quad p(\Sigma) \propto |\Sigma|^{-\frac{1}{2}(p+1)}.$$

From (5.3) it follows that the marginal posterior density function for Σ is the "inverted" Wishart distribution.

$$(5.4) \quad p(\Sigma | \text{data}) \propto |\Sigma|^{-\frac{v}{2}} \exp(-\frac{1}{2} \text{tr} \Sigma^{-1} S),$$

with $v = n - q + p$ and S is the residual sum of squares and product matrix.

Although it is not possible to derive the exact moments of ξ analytically, the distribution of ξ as well as the moments can be obtained by using Monte Carlo simulation.

The simulation procedure follows the same lines as those given in section 4 for the known variance case, except that an extra step is needed at the start where Σ^{-1} is drawn from a $W(S^{-1}, n - k)$ distribution. The histogram of 10 000 simulated values of ξ , the Pearson curve and the Cornish-Fisher approximation of the distribution of ξ are given in figure 5.1.

Prediction of ξ was again based on the same six responses as given in (4.9) but Σ was considered unknown, and drawn from an inverted Wishart distribution (equation (5.4)). The Cornish-Fisher and Pearson approximations were obtained, using the simulated moments.

Posterior distribution of ξ - Σ unknown

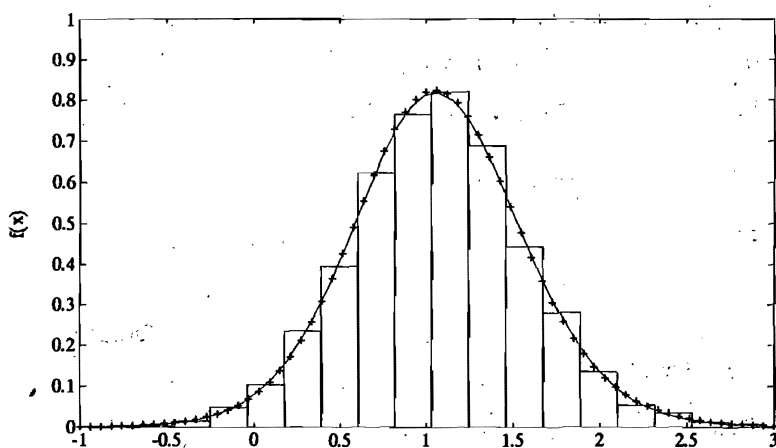


Figure 5.1 Histogram of 10 000 simulated values of ξ
 — Pearson Curve Approximation
 ***** Cornish-Fisher Approximation

Inspection of figure 5.1 shows that the graph for Σ unknown is similar to the graph when Σ is known, but the peak is shorter. The Pearson and Cornish-Fisher densities, as in the case of the Σ -known case, fit the "exact" distribution quite well. Table 5.1 gives the 90%, 95% and 99% "exact" credibility intervals for the Σ -unknown case.

Table 5.1 Credibility intervals

90%	95%	99%
0,2405-1,9068	0,0761-2,0842	-0,2374-2,5166

As mentioned earlier, the main reason for doing this research was to extend the procedure of Hunter and Lamboy to multivariate calibration and to determine whether it differs much from the classical procedure. For comparison purposes the same set of observations 2, 5, 11, 16, 18, 22, 28, 30 and 35 for the paint data as given in Brown (1982) was used for prediction. The nine resulting Y 's (using only responses Y_1, Y_2) were to be used to predict the corresponding ξ (viscosity values). As in section 4 the remaining 27 observations were used to estimate the relationship between X and Y .

Table 5.2 gives the true ξ values for the nine observations, the predictions $\hat{\xi}$ (the classical estimator, equation (3.5)), $\text{Med}(\xi)$ and $\hat{E}(\xi)$ (Hunter and Lamboy's procedures) as well as the percentage of unexplained variation aggregated over the nine observations for the two procedures. $\text{Med}(\xi)$ is the median of ξ , while $\hat{E}(\xi)$ is the arithmetic mean, obtained from 1000 simulations for each of the 9 observations. The 95% confidence intervals of ξ in the case of the classical estimator as well as the 95% credibility intervals using Hunter and Lamboy's procedure are also given.

Inspection of table 5.2 shows that there is not much to choose between the two procedures. $\text{Med}(\xi)$ and $\hat{E}(\xi)$ were closer to the true ξ than $\hat{\xi}$ in five of the nine cases, while the percentage of unexplained variation were 27,25% and 27,86% compared to 27,89% in favour of Hunter and Lamboy's procedure. From the table it also follows that the 95% credibility intervals using Hunter and Lamboy's procedure are substantially shorter than the classical intervals. They are in fact 21,95% shorter while containing the true values with the same frequency as the ordinary intervals.

Table 5.2 Comparison of the two procedures

Classical procedure			Hunter and Lamboy's procedure		
95% Confidence			95% Credibility		
ξ	$\hat{\xi}$	Intervals	$\hat{E}(\xi)$	$\text{Med}(\hat{\xi})$	Intervals
0,00	-0,209	-1,629 - 0,995	-0,204	-0,199	-0,174 - 0,742
1,00	0,371	-0,942 - 1,571	0,380	0,368	-0,598 - 1,348
2,00	1,799	0,598 - 3,143	1,794	-1,768	0,752 - 2,996
0,00	0,666	-0,659 - 1,930	0,668	0,680	-0,279 - 1,642
1,00	1,172	0,030 - 2,405	1,179	1,164	0,210 - 2,176
2,00	2,618	1,389 - 4,140	2,629	2,583	1,631 - 3,935
0,00	0,117	-1,166 - 1,228	0,120	0,136	-0,852 - 1,072
1,00	0,675	-0,513 - 1,804	0,685	0,685	-0,311 - 1,680
2,00	2,467	1,352 - 3,845	2,465	2,443	1,376 - 3,803
Percentage of unexplained variation					
27,89%			27,86%	27,25%	

APPENDIX A

A.1 The following lemmas were used in the proof of theorem 3.1

LEMMA A.1

Suppose $Z \sim N(\tau, I_p)$, then $E(\frac{1}{Z'Z}) = E(W^{-1}) = E(\frac{1}{2K+p-2})$ where

$$(A.1) \quad E(\frac{1}{2K+p-2}) =$$

$$\left\{ \begin{aligned} & \frac{1}{2} \left\{ \sum_{t=1}^{p-1} \frac{(-2)^t}{\lambda^t} \frac{(\frac{1}{2}p-2)!}{(\frac{1}{2}p-t-1)!} \frac{(-2)^{\frac{1}{2}p-1}}{\lambda^{\frac{1}{2}p-1}} e^{-\frac{1}{2}\lambda} (\frac{1}{2}p-2)! \right\}, \text{ for } p \text{ even} \\ & \frac{1}{2} \left\{ \frac{2}{\lambda} + \sum_{t=2}^{\frac{1}{2}(p-3)} \frac{(-2)^{t+1}}{\lambda^t} \prod_{i=2}^t (\frac{1}{2}p-i) \right\} + \frac{2(-2)^{\frac{1}{2}(p-3)}}{\lambda^{\frac{1}{2}(p-2)}} D(\sqrt{\lambda}) \prod_{i=1}^{\frac{1}{2}(p-1)} (\frac{1}{2}p-i), \text{ for } p \text{ odd} \end{aligned} \right.$$

$D(\sqrt{\lambda}) = e^{-\frac{1}{2}\lambda} \int_0^{\sqrt{\frac{1}{2}\lambda}} e^{y^2} dy$ is Dawson's integral which is tabulated in Abramowitz and Stegun (1965, p. 319).

PROOF: The proof of equation (A.1) is given in Egerton and Laycock (1982).

COROLLARY: By applying the recurrence relationship given in Egerton and Laycock (1982) it follows that

$$(A.2) \quad E\left(\frac{1}{2K+p-4}\right) = \left(\frac{1}{p-4}\right)\{1 - \lambda E\left(\frac{1}{2K+p-2}\right)\}.$$

Expressions for $E\left(\frac{1}{2K+p}\right)$, $E\left(\frac{1}{2K+p+2}\right)$, $E\left(\frac{1}{2K+p+4}\right)$ and $E\left(\frac{1}{2K+p+6}\right)$ in terms of $E\left(\frac{1}{2K+p-2}\right)$ can be obtained in a similar way.

The mean, variance and the third moment of ξ exist only for $p > 2$ because $E\left(\frac{1}{2K+p-2}\right)$ exists only for $p > 2$. Since $E\left(\frac{1}{2K+p-4}\right)$ occurs in the first term of (3.4) it also follows that the fourth moment of ξ exists only for $p > 4$.

LEMMA A.2

Since the conditional distribution of $(\eta - \beta_0)$ given β_1 is normal with mean $(y_0 - \bar{y}) + \bar{x}\beta_1$ and covariance matrix $(1 + \frac{1}{n})\Sigma$, expressions for $E\{(\eta - \beta_0)' \Sigma^{-1} \beta_1 | \beta_1\}$, $E\{(\eta - \beta_0)' \Sigma^{-1} \beta_1 | \beta_1\}^2$, $E\{(\eta - \beta_0)' \Sigma^{-1} \beta_1 | \beta_1\}^3$ and $E\{(\eta - \beta_0)' \Sigma^{-1} \beta_1 | \beta_1\}^4$ can easily be obtained. For example

$$(A.3) \quad E\{(\eta - \beta_0)' \Sigma^{-1} \beta_1 | \beta_1\}^2 = (1 + \frac{1}{n})\beta_1' \Sigma^{-1} \beta_1 + \{(y_0 - \bar{y}) + \bar{x}\beta_1\}' \Sigma^{-1} \beta_1\}^2.$$

PROOF: The proof of (A.3) follows easily from expected values of multivariate normally distributed random variables.

LEMMA A.3

Suppose $\mathbf{Z} = (Z_1, Z_2, \dots, Z_p) \sim N(\tau, \mathbf{I}_p)$ and define $\mathbf{Z}'\mathbf{Z} = \mathbf{W}$. Expressions for the expected values $EZ_i W^{-r}$, $EZ_i^2 W^{-r}$, $EZ_i Z_j W^{-r}$, $EZ_i^3 W^{-r}$, $EZ_i^2 Z_j W^{-r}$, $EZ_i Z_j Z_k W^{-r}$, $EZ_i^4 W^{-r}$, $EZ_i^3 Z_j W^{-r}$, $EZ_i^2 Z_j^2 W^{-r}$, $EZ_i^2 Z_j Z_k W^{-r}$ and $EZ_i Z_j Z_k Z_l W^{-r}$ can be obtained. For example

$$(A.4) \quad EZ_i W^{-r} = E(Z_i - \tau_i) W^{-r} + \tau_i E W^{-r} \\ = \frac{\partial}{\partial \tau_i} E W^{-r} + \tau_i E W^{-r}$$

$$(A.5) \quad EZ_i^2 W^{-r} = E(Z_i - \tau_i)^2 W^{-r} + 2\tau_i E(Z_i - \tau_i) W^{-r} + \tau_i^2 E W^{-r} \\ = \left[\frac{\partial^2}{\partial \tau_i^2} + 1 \right] E W^{-r} + 2\tau_i \frac{\partial}{\partial \tau_i} E W^{-r} + \tau_i^2 E W^{-r}$$

and

$$(A.6) \quad EZ_i Z_j W^{-r} = E(Z_i - \tau_i)(Z_j - \tau_j) W^{-r} + \tau_i E(Z_j - \tau_j) W^{-r} + \tau_j E(Z_i - \tau_i) W^{-r} \\ + \tau_i \tau_j E W^{-r} \\ = \frac{\partial^2}{\partial \tau_i \partial \tau_j} E W^{-r} + \tau_i \frac{\partial}{\partial \tau_j} E W^{-r} + \tau_j \frac{\partial}{\partial \tau_i} E W^{-r} + \tau_i \tau_j E W^{-r}$$

PROOF: By extending the results of Ullah (1974, equations (17) - (20)) the lemma follows.

LEMMA A.4

The derivatives $\frac{\partial}{\partial \tau_i} E W^{-r}$, $\frac{\partial^2}{\partial \tau_i^2} E W^{-r}$, $\frac{\partial^3}{\partial \tau_i^3} E W^{-r}$, $\frac{\partial^3}{\partial \tau_i^3} E W^{-r}$, $\frac{\partial^4}{\partial \tau_i^4} E W^{-r}$, $\frac{\partial^2}{\partial \tau_i \partial \tau_j} E W^{-r}$, $\frac{\partial^3}{\partial \tau_i^2 \partial \tau_j} E W^{-r}$, $\frac{\partial^4}{\partial \tau_i^2 \partial \tau_j^2} E W^{-r}$, $\frac{\partial^3}{\partial \tau_i \partial \tau_j \partial \tau_k} E W^{-r}$, $\frac{\partial^4}{\partial \tau_i^2 \partial \tau_j \partial \tau_k} E W^{-r}$ and $\frac{\partial^4}{\partial \tau_i \partial \tau_j \partial \tau_k \partial \tau_l} E W^{-r}$ can be obtained in a straight-forward manner. For

example

$$(A.7) \quad \frac{\partial}{\partial \tau_i} EW^{-r} = \left[\frac{\partial}{\partial \bar{\lambda}} EW^{-r} \right] \frac{\partial \bar{\lambda}}{\partial \tau_i} = \left[\frac{\partial}{\partial \bar{\lambda}} EW^{-r} \right] \tau_i$$

$$(A.8) \quad \frac{\partial^2}{\partial \tau_i^2} EW^{-r} = \left[\frac{\partial^2}{\partial \bar{\lambda}^2} EW^{-r} \right] \tau_i^2 + \frac{\partial}{\partial \bar{\lambda}} EW^{-r}$$

$$(A.9) \quad \frac{\partial^2}{\partial \tau_i \partial \tau_j} EW^{-r} = \left[\frac{\partial^2}{\partial \bar{\lambda}^2} EW^{-r} \right] \tau_i \tau_j$$

PROOF: The results follow from straight-forward differentiation. See also Ullah (1974, equations (27) and (28)).

LEMMA A.5

By making use of the fact that

$$(A.10) \quad \frac{\partial^s}{\partial \bar{\lambda}^s} EW^{-r} = 2^{-r} (-1)^s \frac{\Gamma(r+s) \Gamma(\frac{p}{2} - r)}{\Gamma(r) \Gamma(\frac{p}{2} + s)} e^{-\bar{\lambda}} {}_1F_1(\frac{p}{2} - r; \frac{p}{2} + s; \bar{\lambda})$$

where ${}_1F_1(a; c; x) = \sum_{l=0}^{\infty} \frac{(a)_l}{(c)_l} \frac{x^l}{l!}$, $(a)_l = a(a+1) \dots (a+l-1)$ and $a_0 = 1$, the

expected values EW^{-1} , EW^{-2} , $\frac{\partial}{\partial \bar{\lambda}} EW^{-1}$, $\frac{\partial}{\partial \bar{\lambda}} EW^{-2}$, $\frac{\partial^2}{\partial \bar{\lambda}^2} EW^{-1}$ and $\frac{\partial^2}{\partial \bar{\lambda}^2} EW^{-2}$ can be written in terms of expected values of Poisson random variables. For example

$$(A.11) \quad EW^{-1} = E\left(\frac{1}{p+2K-2}\right)$$

$$(A.12) \quad EW^{-2} = E\left\{\frac{1}{(p+2K-2)(p+2K-4)}\right\} = \frac{1}{2} E\left\{\frac{1}{(p+2K-4)} - \frac{1}{(p+2K-2)}\right\}$$

$$(A.13) \quad \frac{\partial}{\partial \lambda} E W^{-2} = -4E \left\{ \frac{1}{(p+2K)(p+2K-2)(p+2K-4)} \right\} \\ = -\frac{1}{2}E \left\{ \frac{1}{(p+2K)} - \frac{2}{(p+2K-2)} + \frac{1}{(p+2K-4)} \right\}$$

$$(A.14) \quad \frac{\partial^2}{\partial \lambda^2} E W^{-2} = 24E \left\{ \frac{1}{(p+2K+2)(p+2K)(p+2K-2)(p+2K-4)} \right\} \\ = \frac{1}{2}E \left\{ \frac{1}{(p+2K-4)} - \frac{3}{(p+2K-2)} + \frac{3}{(p+2K)} - \frac{1}{(p+2K+2)} \right\}$$

PROOF:

Equation (A.10) is given in Slater (1960, p. 15, equation (2.1.5)) and Ullah (1974, p. 147, equation (26)) and follows from the fact that $W \sim \chi_p^2(\lambda)$. By substituting for example $s = 1$ and $r = 2$ in (A.10) and using partial fractions, (A.14) follows. The derivatives for other combinations of r and s can be obtained in a similar way.

LEMMA A.6

For $h(p \times 1)$ and $Z(p \times 1) \sim N(\tau, I_p)$ it follows that

$$(A.15) \quad W(h'Z)W^{-r} = E \sum_{i=1}^p h_i Z_i W^{-r}$$

$$(A.16) \quad E(h'Z)^2 W^{-r} = E \left\{ \sum_{i=1}^p h_i^2 Z_i^2 + \sum_{i=1}^p \sum_{j=1}^p h_i h_j Z_i Z_j \right\} W^{-r}$$

PROOF:

The proof of (A.16) is as follows:

$$E(h'Z)^2 W^{-r} = E \left\{ \left[\sum_{i=1}^p h_i Z_i \right]^2 W^{-r} \right\} \\ = E(h_1 Z_1 + h_2 Z_2 + \dots + h_p Z_p)^2 W^{-r} \\ = E\{(h_1 Z_1)^2 + \dots + (h_p Z_p)^2 + 2(h_1 Z_1)(h_2 Z_2) + \dots \\ + 2(h_{p-1} Z_{p-1})(h_p Z_p)\} W^{-r} \\ = E \left\{ \sum_{i=1}^p h_i^2 Z_i^2 + \sum_{i=1}^p \sum_{j=1}^p h_i h_j Z_i Z_j \right\} W^{-r}$$

A.2 Proof of theorem 3.1

Only the proof of $E(\xi^2)$ will be given

$$E(\xi^2) = E(E(\xi^2 | \beta_1)) = E \left\{ E \left[\frac{(\eta - \beta_0)' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1} \right]^2 | \beta_1 \right\}.$$

First consider

$$(A.17) \quad E((\eta - \beta_0)' \Sigma^{-1} \beta_1 | \beta_1)^2.$$

By using (A.3) in lemma A.2, it follows from equation (A.17) that

$$(A.18) \quad E(\xi^2) = \left(1 + \frac{1}{n}\right) E \left\{ \frac{1}{\beta_1' \Sigma^{-1} \beta_1} \right\} + E \left\{ \frac{(y_0 - \bar{y})' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1} \right\}^2 \\ + 2\bar{x} E \left\{ \frac{(y_0 - \bar{y})' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1} \right\} + \bar{x}^2.$$

From equation (A.18), first consider

$$(A.19) \quad \left(1 + \frac{1}{n}\right) E \left\{ \frac{1}{\beta_1' \Sigma^{-1} \beta_1} \right\} = \left(1 + \frac{1}{n}\right) E \left\{ \frac{C}{C \beta_1' \Sigma^{-1} \beta_1} \right\} \\ = \left(1 + \frac{1}{n}\right) CE \left\{ \frac{1}{Z'Z} \right\} \\ = \left(1 + \frac{1}{n}\right) CE (W^{-1}) \\ = \left(1 + \frac{1}{n}\right) CE \left(\frac{1}{(p+2K-2)} \right)$$

where $C = \bar{C}^{-1} = \sum_{i=1}^n (x_i - \bar{x})^2$; $\beta_1 \sim N(\bar{\beta}_1, \bar{C}\Sigma)$;

$Z = C^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \beta_1 \sim N(\tau, I_p)$; $\tau = C^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \bar{\beta}_1$ and where K is a Poisson

random variable with parameter $\bar{\lambda} = \frac{1}{2} \lambda \tau' \tau$.

Secondly, consider $E \left\{ \frac{(\mathbf{y}_0 - \bar{\mathbf{y}})' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1} \right\}^2$ in equation (A.18).

$$\begin{aligned}
 E \left\{ \frac{(\mathbf{y}_0 - \bar{\mathbf{y}})' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1} \right\} &= CE \left\{ \frac{(\mathbf{h}' \mathbf{Z})^2}{(\mathbf{Z}' \mathbf{Z})^2} \right\} \\
 &= CE (\mathbf{h}' \mathbf{Z})^2 W^{-2} \\
 (A.20) \quad &= CE \left\{ \sum_{i=1}^p h_i^2 Z_i^2 + \sum_{i=1}^p \sum_{j=1}^p h_i h_j Z_i Z_j \right\} W^{-2}, \\
 &\quad \text{[using equation (A.16), lemma A.6]}
 \end{aligned}$$

where $\mathbf{h} = (\mathbf{y}_0 - \bar{\mathbf{y}})' \Sigma^{-1/2}$.

Using lemma A.3, equations (A.4) and (A.6) as well as lemma A.4 equations (A.7) - (A.9) and (for $r=2$), it can be shown that equation (A.20) is equal to

$$\begin{aligned}
 &\left\{ \sum_{i=1}^p h_i^2 \tau_i^2 + \sum_{i=1}^p \sum_{j=1}^p h_i h_j \tau_i \tau_j \right\} \left[\frac{\partial^2}{\partial \lambda^2} E(W^{-2}) \right] \\
 &+ \left\{ 2 \left[\sum_{i=1}^p h_i^2 \tau_i^2 + \sum_{i=1}^p \sum_{j=1}^p h_i h_j \tau_i \tau_j \right] + \sum_{i=1}^p h_i^2 \right\} \left[\frac{\partial}{\partial \lambda} E(W^{-2}) \right] \\
 &+ \left\{ \left[\sum_{i=1}^p h_i^2 \tau_i^2 + \sum_{i=1}^p \sum_{j=1}^p h_i h_j \tau_i \tau_j \right] + \sum_{i=1}^p h_i^2 \right\} E(W^{-2}) \\
 &= (\mathbf{h}' \boldsymbol{\tau})^2 \left[\frac{\partial^2}{\partial \lambda^2} E(W^{-2}) \right] + [2(\mathbf{h}' \boldsymbol{\tau})^2 + (\mathbf{h}' \mathbf{h})] \left[\frac{\partial}{\partial \lambda} E(W^{-2}) \right] \\
 &+ [(\mathbf{h}' \boldsymbol{\tau})^2 + (\mathbf{h}' \mathbf{h})] E(W^{-2}).
 \end{aligned}$$

Using lemma A.5 equations (A.11) - (A.14) it follows that

$$\begin{aligned}
 (A.21) \quad CE(\mathbf{h}' \mathbf{Z})^2 W^{-2} &= C \left\{ \frac{1}{2} (\mathbf{h}' \mathbf{h}) E \left(\frac{1}{p+2K-2} \right) + \left[\frac{1}{2} (\mathbf{h}' \boldsymbol{\tau})^2 - \frac{1}{2} (\mathbf{h}' \mathbf{h}) \right] E \left(\frac{1}{p+2K} \right) \right. \\
 &\quad \left. - \frac{1}{2} (\mathbf{h}' \boldsymbol{\tau})^2 E \left(\frac{1}{p+2K+2} \right) \right\}.
 \end{aligned}$$

Thirdly, consider $E \left\{ \frac{(\mathbf{y}_0 - \bar{\mathbf{y}})' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1} \right\}$ in equation (A.18)

$$\begin{aligned}
 2\bar{x}E \left\{ \frac{(\mathbf{y}_0 - \bar{\mathbf{y}})' \Sigma^{-1} \beta_1}{\beta_1' \Sigma^{-1} \beta_1} \right\} &= 2\bar{x}C^{\frac{1}{2}} E \left\{ \frac{\mathbf{h}' \mathbf{Z}}{\mathbf{Z}' \mathbf{Z}} \right\} = C^{\frac{1}{2}} E \left\{ \left[\sum_{i=1}^p h_i \mathbf{Z}_i \right] \mathbf{W}^{-1} \right\} \\
 &= 2\bar{x}C \sum_{i=1}^p h_i E(\mathbf{Z}_i \mathbf{W}^{-1}) \\
 (A.22) \quad &= \tilde{C}^{-\frac{1}{2}} (\mathbf{h}' \boldsymbol{\tau}) E \left(\frac{1}{p+2K} \right) \text{ from equations (A.15), (A.4)} \\
 &\quad \text{and (A.7).}
 \end{aligned}$$

Substituting (A.19), (A.21) and (A.22) into (A.18), the proof follows. For further details about lemmas A.1–A.6 and theorem 3.1, see Du Plessis and Van der Merwe (1991).

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